

Experimental Demonstration of Polar Coded Many-to-one Mapping based Truncated PS-PAM8 with 37-dB power budget

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Abstract *In this paper, a polar coded truncated probabilistic shaped PAM-8 (PTPS-PAM8) operating at 200-Gb/s/lane is proposed and experimentally investigated using the intensity-modulation and coherent-detection structure. Adopting the PTPS-PAM8, a 2-dB ROP sensitivity improvement is obtained over the uniformly-distributed polar-coded PAM8 after 20-km transmission. ©2026 The Author(s)*

Introduction

Passive optical network (PON) technology has attracted tremendous attention due to its powerful ability to provide high-quality optical access services at low cost, which always adopts the intensity-modulation and direct-detection (IM-DD) structure for several years [1]. Currently, the ITU-T is working on the development of the Very High Speed PON (VHSP) standard, and 200G is likely to be the successor to 50G PON [2]. However, challenges induced by limited bandwidth need to be addressed when employing the IM-DD technique to scale up the data rate to 200Gb/s [3], [4]. Hence, to achieve 200Gb/s based VHSP, high-order intensity-modulation formats might be necessary to relax the bandwidth requirements of electrical components. Nevertheless, increasing the modulation order would deteriorate the tolerance to system impairments inevitably, necessitating a higher received optical power (ROP) [5]. As a result, poor power budget performance is obtained. To address this issue, truncated probabilistic shaping (TPS) techniques might be a promising approach, which can enhance system performance by generating non-uniformly distributed symbols and forcing the probabilities of some high-power points to zero simultaneously. Its strong potential in delivering superior performance has been widely investigated in quadrature amplitude modulation (QAM)-based systems [6]. However, its effectiveness in enhancing the performance of intensity modulation (IM)-based systems has not yet been explored, such as the IM-based PON systems.

PON is a cost-sensitive point-to-multi-point system, which consists of one optical line terminal (OLT) and multiple optical network units (ONUs). Hence, each ONU bears its own costs and shares the cost of OLT with other ONUs

connected with the same OLT. Hence, when using the TPS techniques in IM-based PON systems, it is essential to thoroughly investigate its low-cost implementations, especially for the upstream direction where signals are launched from the ONU and received by the OLT. To date, the reported schemes for achieving TPS can be roughly classified into two types, i.e., distribution matcher (DM) based schemes and many-to-one (MTO) mapping-based schemes [7]. In terms of the DM-based schemes, an extra distribution matcher is required with some additional complexity. Therefore, reducing this complexity is crucial when deploying DM-based TPS in PONs. In contrast, MTO-based schemes implement TPS by modifying the bit-to-symbol mapping rules of the existed symbol mapper, eliminating the need for extra DM modules. Hence, the complexity of MTO-based TPS schemes could be lower than that of DM based schemes. From a cost perspective, the MTO schemes might be suitable candidates, better aligned with the low-cost requirements of PON systems.

Additionally, aiming to enhance the VHSP performance, the effects induced by the combination of chromatic dispersion and direct detection (DD) receiver are also worthy of considerations [8]. For double sideband signals, the two sidebands experience different phase variations due to the chromatic dispersion. Thereby, significant power fading effects on the spectrum are observed when employing the DD receiver to detect IM signals, limiting the sensitivity and power budget metrics. To alleviate this bottleneck, replacing the DD receiver with a coherent detection receiver is a viable option as it can detect the intensity and phase information simultaneously, contributing to enhanced system performance [3].

Therefore, in this paper, a 50-GBaud dual-polarized polar coded many-to-one (MTO) mapping

based truncated probabilistic shaped 8-ary pulse amplitude modulation (PTPS-PAM8) is proposed and investigated over a 20-km PON upstream link, deploying the intensity modulation at the optical network unit (ONU) side and coherent detection at optical line terminal (OLT) side respectively. Thereby, the low-cost of ONU could be maintained while the ROP sensitivity could also be improved. Experimental results show that a 37-dB power budget is obtained by the proposed PTPS-PAM8. Meanwhile, 2-dB power budget improvement could be introduced when comparing to the polar coded uniformly-distributed PAM8 (PUD-PAM8) over 20-km fibre.

Principle of PTPS-PAM8

The principle of PTPS-PAM8 is shown in Fig. 1 (a). A systematic polar encoder is employed first to indicate information bit positions directly [7]. The code length and code rate are $N=256$ and $R=2/3$ respectively. Then, the polar coded binary streams are interleaved by a bit-wise interleaver and cascaded by an MTO mapper to output PTPS-PAM8 signals, whose mapping rules are demonstrated in Fig. 1(b). As shown, the least significant bit (LSB) is provided by the frozen bits of polar codewords, and the other two bits are obtained from the information bits of polar codewords. Finally, 3-bit sequences are mapped to five-level signals. For instance, both bit sets "111" and "110" are mapped to symbol "0", and the LSB of symbol "0" labelled by "X" denotes that it could be either bit "0" or bit "1". Hence, such an LSB is ambiguous for the polar decoder, which would be recovered by the puncturing-aided polar decoding algorithms [7]. It should be mentioned that puncturing operations may degrade the error-correcting abilities of polar code within a limited range as investigated in Fig. 1(c). Herein, owing to the limitation that low BER values could not be reliably estimated in simulations, such as $1E-15$, an outer low-complexity code having a pre-BER threshold of $1E-3$ to achieve a post-BER of $1E-15$ is assumed here [9]. As shown in Fig. 1(c), the pre-BER thresholds of PTPS-PAM8 and PUD-PAM8 are approximately $1.5E-2$ and $6.3E-2$ respectively when using the puncturing aided successive cancellation list (SCL) polar decoding algorithm with a list size of $L=32$. Notably, although the error-correcting abilities of polar code in PTPS-PAM8 are a little weakened compared to that of PUD-PAM8, PTPS-PAM8 could still introduce performance improvements compared to PUD-PAM8 as experimental investigations appended below due to its significant pre-BER superiority.

Experimental Setup and Results Discussions

The experimental setup with the ONU and OLT side DSP is demonstrated in Fig. 2. At the ONU-side, two pseudo random bit sequences (PRBS)

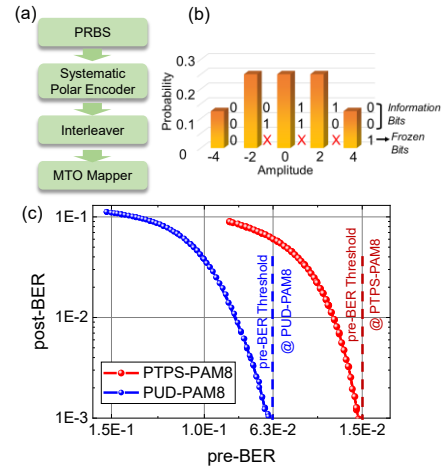


Fig. 1. (a) The schematic diagram of PTPS-PAM8 and (b) the mapping rules of MTO mapper. (c) The post-BER versus pre-BER.

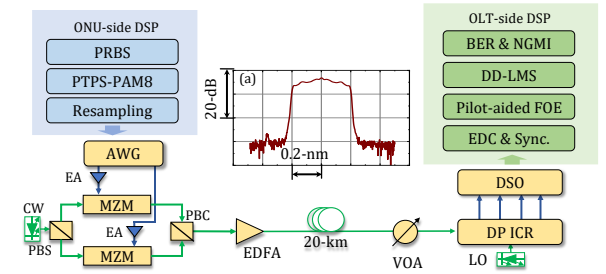


Fig. 2: The experimental setup. (a) The optical spectrum of PTPS-PAM8.

with different lengths are generated and utilized to generate double-polarized PTPS-PAM8 (DP-PTPS-PAM8) symbols. Then, the DP-PTPS-PAM8 symbols are resampled and fed to the arbitrary waveform generator (AWG, Keysight M8196A) operating at 92-GSa/s to generate 50-GBaud electrical signals. A DP IQ modulator (Lumentum, CDM) biased at the null point is used to generate DP-PTPS-PAM8 optical signals [10]. Thus, only one branch in each polarization of the DP IQ modulator would be used. The continuous waveform (CW) of the DP IQ modulator at 1550-nm is provided by a tunable laser source (Keysight N7714A) with a linewidth of 100-kHz. Before coupling the signals into a 20-km fibre, an erbium-doped fibre amplifier (EDFA) is inserted to boost the launch power of PTPS-PAM8 to approximately 9-dBm. Following the methods mentioned in [7], the net data rate of PTPS-PAM8 is 200-Gb/s ($50\text{-GBaud} \times 3\text{-bit/symbol} \times 2/3\text{-code rate} \times 2\text{-polarizations}=200\text{-Gb/s}$). In addition, a 50-GBaud DP-PUD-PAM8 with the same polar code length $N=256$ and code rate $R=2/3$ is also used, whose net data rate is also 200-Gb/s.

After 20-km transmission, a variable optical attenuator (VOA) is utilized to adjust the received optical power (ROP). Then, the signals are detected by an integrated coherent receiver (ICR, Fujitsu FIM24706/301). Afterwards, the signals are digitized by an 80-GSa/s digital sampling

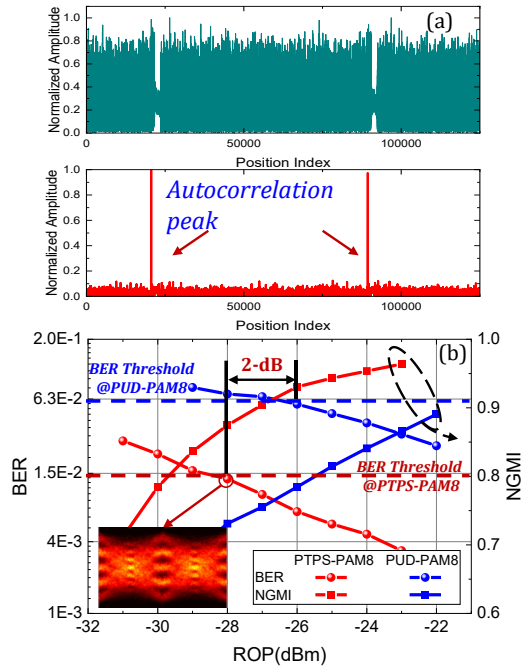


Fig. 3. (a) The normalized auto-correlation peak in frame synchronization. (b) The BER and NGMI versus ROP.

oscilloscope (DSO) and processed by offline DSP as shown in Fig. 2, including electrical chromatic dispersion compensation (EDC), pilot-aided frame synchronization, pilot-aided frequency offset estimation (FOE) and pilot-aided decision-directed least mean square (DD-LMS) based adaptive equalization. Finally, the bit error ratio (BER) measurement and normalized generalized mutual information (NGMI) are estimated.

Figure 3(a) shows that the autocorrelation peak of frame synchronization could be successfully detected with the pilot length of 512, introducing a time duration of approximately 10.25-ns. Fig. 3(b) demonstrates the BER and NGMI under optical back-to-back (OBTB) case. As shown, the ROP sensitivities of PTPS-PAM8 and PUD-PAM8 are -28dBm and -26dBm respectively, suggesting that 2-dB ROP sensitivity improvement could be achieved by PTPS-PAM8 compared to PUD-PAM8. Considering the launch power is 9-dBm, the power budget of PTPS-PAM8 and PUD-PAM8 would reach 37-dB and 35-dB respectively. Besides, the NGMI metrics also verifies that the PTPS-PAM8 could outperforms PUD-PAM8 over the measured ROP range.

Then, system performance is investigated using 20-km standard single mode fibre (SSMF). As depicted in Fig. 4 (a), the performance would be affected by launch power significantly, and the optimal launch powers of PTPS-PAM8 and PUD-PAM8 are both 9-dBm. Therefore, the BER and NGMI performance over 20-km fibre are investigated with the launch power of 9-dBm as shown in Fig. 4(b). Results show that the ROP sensitivity of PTPS-PAM8 over 20-km fibre transmission is

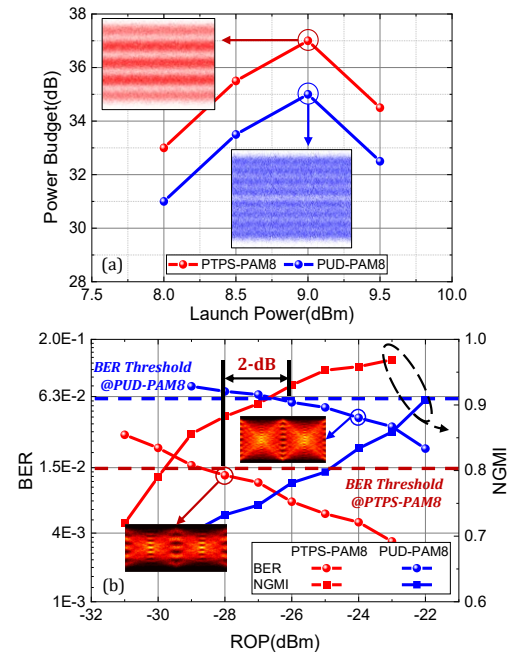


Fig. 4. (a) The power budget versus launch power over 20-km fibre. (b) The BER and NGMI versus ROP.

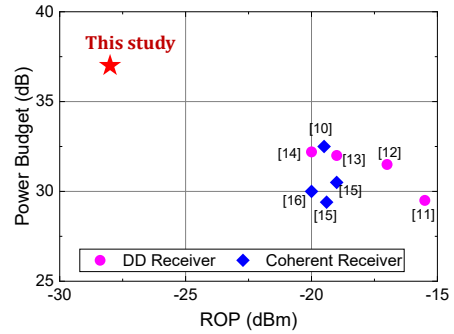


Fig. 5. Recent studies regarding intensity-modulation based 200-Gb/s/λ PON using SSMF.

-28dBm, contributing to a power budget of 37-dB. In terms of PUD-PAM8, the ROP sensitivity would be deteriorated to about -26dBm, resulting in a decreased power budget of 35-dB. Results over 20-km transmission further confirms the superiority of PTPS-PAM8 in achieving better performance compared with PUD-PAM8.

Finally, recent studies on IM-based 200Gb/s/λ PON using SSMF are summarized in Fig. 5 [10-16]. The BER threshold of 1E-2 is utilized for fair comparisons. As shown, a remarkable ROP sensitivity and power budget is achieved in this paper.

Conclusions

In this paper, a PTPS-PAM8 is proposed and experimentally investigated over a 20-km upstream PON system, which demonstrates strong abilities in enhancing ROP sensitivity and improving power budget metrics comparing with that of PUD-PAM8. Employing this novel scheme, a 200-Gb/s upstream PON is demonstrated with remarkable ROP sensitivity and power budget of -28dBm and 37-dB respectively.

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